Justin Jaffeth Corea Masís

Electronic Engineering

Location: Cartago, Costa Rica (Available for Relocation)
Email: coreajustin510@gmail.com
GitHub: https://github.com/jcorea510
Tel: +506 6117-8713
LinkedIn: https://linkedin.com/in/justincorea
Portfolio: https://jcorea510.github.io/

Profile

Electronic Engineer focused on embedded systems and machine learning. With knowledge in C/C++, Rust, and Python development, as well as Linux-based platforms. Seeking an opportunity to establish myself in a company where I can apply the knowledge acquired during my academic training while continuing to grow professionally.

Projects

Automatic License Plate Recognition System *Final Graduation Project (2025)*

- Designed and implemented an end-to-end computer vision pipeline (detection + OCR) for license plate recognition, targeting deployment in embedded systems for automated parking applications.
- Trained models using synthetically generated data to address dataset limitations, and validated performance on real-world data.
- Achieved 98.44% accuracy in plate recognition and 93.5% detection accuracy.
- Reached 3.84 FPS inference speed on resource-constrained hardware.

Networked Sensor Node *Data Science (2026)*

- Developed firmware in MicroPython for Raspberry Pi Pico W sensor nodes.
- Designed a lightweight TCP communication protocol for automatic node registration and telemetry.
- Implemented end-to-end data acquisition system integrated with Python server and MySQL database.

STM32-Based Oscilloscope *Embedded Systems (2026)*

- Developed a low-cost oscilloscope using an STM32F103C8T6 ADC for signal acquisition and measurement.
- Implemented ADC DMA and FreeRTOS to improve real-time sampling and data transfer performance.
- Developed a Rust-based PC interface communicating with the embedded device through UART.

16-bit RISC-V Processor Design (Verilog) *Computer Architecture (2024)*

- Designed and simulated a 16-bit RISC-V microarchitecture using Verilog.
- Implemented pipelined datapath including ALU, register file, and control unit.
- Developed testbenches to verify instruction execution and pipeline behavior.

DC Motor Speed Control System (12V) *Hardware Design (2024)*

- Designed and implemented a closed-loop motor speed control system using analog circuitry and sensor feedback.
- Developed hardware for real-time speed regulation under variable load conditions.
- Integrated user interface including emergency stop, reset, mode selection (manual/automatic), and speed visualization.
- Implemented signal conditioning and control using operational amplifiers and sensor inputs.

Education

Licentiate Degree in Electronic Engineering *Instituto Tecnológico de Costa Rica (2020 – 2026)*
GPA: 87 / 100 | Academic Excellence Scholarship

Technical Skills

Programming Languages: C++, Rust, Python
Machine Learning: PyTorch, Deep Learning, Model Optimization
Embedded: STM32, Raspberry Pi, I2C/UART, UDP/TCP
Hardware: Verilog, Analog Design, Digital Design
Tools & Environments: Git, GDB, GitHub, SQL, Linux, CMake
Other: Driving License B1

Languages

Spanish: Native | **English:** CEFR B2 (Prueba SpeakNow) | **Japanese:** JLPT N5